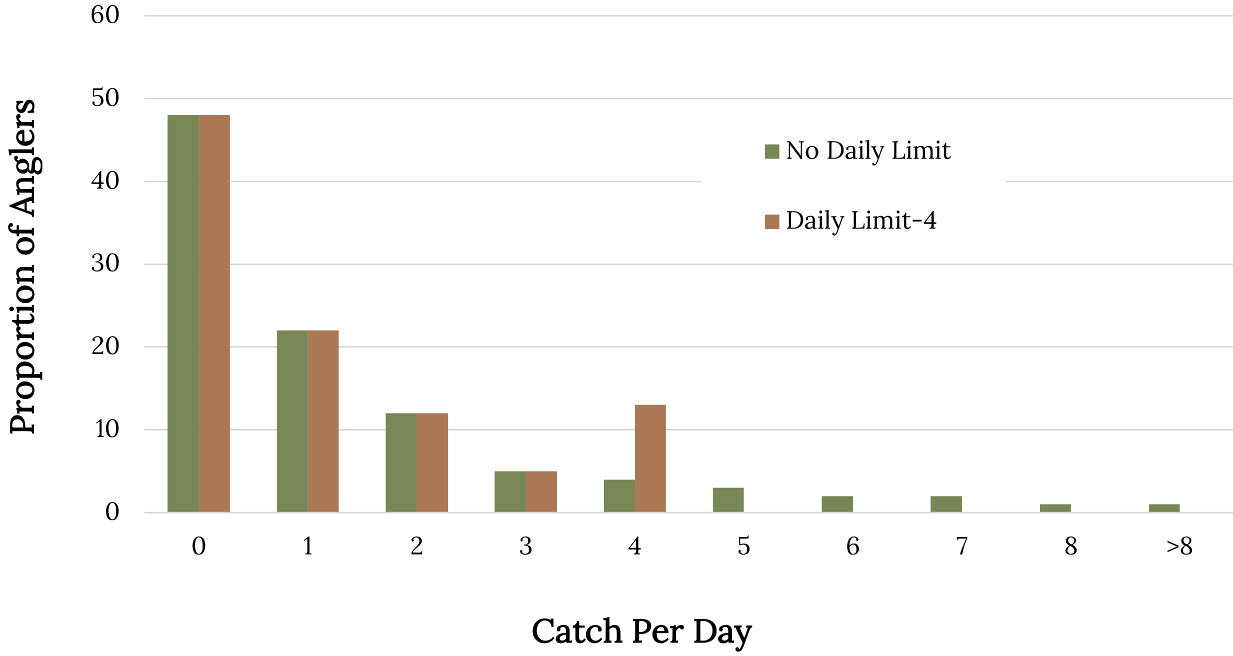

Figure 10.5: Frequency distribution displays the number of angler days resulting in differing catch per day for a hypothetical 8 fish per day creel limit and estimated change if creel limit is reduced to 4 fish per day. [Long description]().

Creel limits are one of many elements that may be used by anglers to define fishing success. When more fish are harvested per trip, anglers rate fishing higher. High creel limits may cause anglers to have unrealistic expectations about the potential supply of fish compared to the demand (Cook et al. 2001). Creel limit reductions may be unsuccessful in reducing angler harvest or affecting fish populations. The hypothetical angler success graph (Figure 10.5) demonstrates that a reduction in creel from 8 to 4 would affect only a few trips and result in a small harvest reduction. Furthermore, creel limits are applied on a per-angler basis, so they cannot control total harvest if total fishing effort increases or if noncompliance is high. Finally, since anglers have a variety of motivations, they likely respond differently to regulation changes (Beard et al. 2011).

The ethic of fairness is involved in setting creel limit regulations because many anglers do not harvest a single fish during an angling trip. In Wisconsin lakes, Walleye harvest was not equally distributed. Only 7.4% of Walleye angler trips were successful in harvesting at least one Walleye, and <1% harvested a limit during a fishing trip (Staggs 1989). In Minnesota, anglers were slightly more successful, where 27.2% of angler trips ended with a harvest of at least one Walleye and about 1% harvesting a limit. The ideal creel limit would distribute the catch among more anglers and prevent overuse by a few individuals.

Long-term trends in panfish populations (i.e., Bluegill, Yellow Perch, Black Crappie, Pumpkinseed, and Rock Bass) in Wisconsin lakes showed significant declines due to overfishing (Rypel et al. 2016). The daily limit for panfish was 50 aggregate per day from 1967 through 1998, which was reduced to 25 in 1998. Further reduction in daily limits for panfish (10) to improve undesirable small sizes of Bluegill populations increased both mean length and mean maximum length relative to sizes in control lakes (Jacobson 2005; Rypel et al. 2015).